

VOCABULARY GROWTH FUNCTIONS

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Key words: Generalised inverse Gaussian-Poisson distribution; statistical linguistics; type-token characteristic; vocabulary.

Summary: This paper outlines techniques developed for use in the field of statistical linguistics – in particular, the modelling of ‘vocabulary growth’ functions, which describe the relationship between the total number of words counted in a text, and either the number of unique words or the number of words of a particular frequency. Many models have been suggested, however, the majority of these suffer from various theoretical and practical drawbacks. The model proposed by Sichel, which is based on a probability theory, includes many previously-used distributions as limiting or special cases. This model allows for estimation of the total active vocabulary of an individual and has been shown to describe real data very well. In the present work, Sichel’s theory is extended and a new algorithm proposed for estimating model parameters in the ‘non-stationary’ case, where an author’s active vocabulary increases as more observations are taken.

1. Introduction

There has been a good deal of work produced concerning the ‘type-token characteristic’ which describes the relationship between the total number of words and the number of distinct words in a literary text. This question arises not only in the field of statistical linguistics but in any situation where a population is partitioned into a number of classes and one wishes to estimate

how many classes exist. The difficulty stems from the fact that it is not possible to observe, for example, the number of words in an author's vocabulary that do not appear in a text.

Yule (1944) published an interesting and insightful discourse on the subject, but a lack of solid theoretical foundation for the field hampered progress from this time. Numerous suggestions as to the form of the word frequency distribution have been put forward; however, most of them display serious shortcomings and have not been applied successfully to real data. Relevant examples are the Waring distribution (*e.g.* Ratkowsky, Halstead and Hantrais, 1980); the Riemann distribution (Zipf, 1932); models based on stochastic processes (*e.g.*, Gani, 1982 and Simon, 1955), and various others. Attempts at modelling the type-token characteristic have been made by (*inter alia*) Brainerd (1972), Mc Neil (1973) and Chen (1989).

The striking features that emerge from almost all this work are that none of the proposed measures or 'characteristics' (apart from that suggested by Yule, 1944) enables one to compare different texts in a manner which is truly independent of text length (despite claims to the contrary), and methods that seem to be applicable to a certain section of the data generally break down when tested over the entire length of the observations. Weitzman (1986) provides a good discussion. See also Price (1992) and Bunge and Fitzpatrick (1993). It was only relatively recently that a promising model based on a probability theory was proposed by Sichel (1971, 1975 and 1986a). It is this model that forms the basis for the present work.

2. Notation, Definitions and Background

In a given text with total number of words N (called *tokens*) there will be a certain number of distinct words (called *types*) which we denote by n_N . This information is embedded in the word probability distribution itself, which gives the number of times unique word types are used in a text count of N word tokens. The probability of any word type appearing $r = 0, 1, 2, \dots, N$ times in a token count of N will be denoted by $\phi(r|N)$ and the corresponding frequency function by $f_{r,N} (= n_N \phi(r|N))$. The relationship between n_N and N (the '*type-token characteristic*') can be written $n_N = \Omega(N)$. Words occurring exactly once in a text are called *hapax legomena*, and $f_{1,N}$ represents the number of *hapax legomena* in a sample of N tokens.

Maas (1972) sets out four requirements that must be adhered to if the logic of a model for n_N is not to be compromised (assuming n_N to be continuous rather than discrete, as is usually done for the sake of convenience): (i) $\lim_{N \rightarrow \infty} n_N$ must be finite; (ii) Since the derivative of the function at any point gives the relative increase in vocabulary, we should have that $\lim_{N \rightarrow \infty} \frac{d}{dN} n_N = 0$; (iii) The function evaluated at the point 1 must be equal to 1; and (iv) $\frac{d}{dN} n_N$ must decrease monotonically from 1 to 0.

Two additional necessary aspects of any candidate model are that it must be capable of the required amount of skewness (this has been a problem with many previously-used distributions) and it must be able to reproduce a mode at any frequency r , since there is no theoretical reason why the mode should be at frequency 1 (see Sichel, 1986b). Sichel's model, which includes many previously-used distributions as limiting or special cases, complies with both

of these conditions, as well as forming the basis for a type-token characteristic that conforms to Maas' specifications (Price, 1992) and allows for estimation of the total active vocabulary of an individual. It thus represents, in all senses, an improvement over much previous work.

3. The Type-Token Characteristic and Related Functions

Sichel (1986b) derives the type-token characteristic

$$n_N = \frac{2}{bc} \cdot \frac{K_\gamma(b)}{K_{\gamma+1}(b)} \left[1 - \frac{K_\gamma(b\sqrt{1+cN})}{(\sqrt{1+cN})^\gamma K_\gamma(b)} \right] \quad (1)$$

and the underlying word frequency distribution, the generalized inverse Gaussian-Poisson (GIGP or 'Sichel') distribution, in its zero-truncated form:

$$\phi(r|N) = \frac{1}{\left[(\sqrt{1+cN})^\gamma K_\gamma(b) - K_\gamma(b\sqrt{1+cN}) \right]} \cdot \frac{1}{r!} \cdot \left(\frac{bcN}{2\sqrt{1+cN}} \right)^r K_{r+\gamma}(b\sqrt{1+cN})$$

for $r = 1, 2, \dots, \infty$ (2)

where $b > 0$, $c > 0$ and $-\infty < \gamma < \infty$; and $K_\nu(z)$ is the modified Bessel function of the second kind¹ of order ν and argument z .

Hence
$$f_{r,N} = n_N \cdot \phi(r|N) = \frac{N^r}{r!} \cdot \left(\frac{bc}{2} \right)^{r-1} \cdot \frac{K_{\gamma+r}(b\sqrt{1+cN})}{(\sqrt{1+cN})^{\gamma+r} K_{\gamma+1}(b)} \quad (3)$$

¹ This is sometimes referred to in the literature as the modified Bessel function of the third kind.

$$\text{and } f_{1,N} = \frac{N}{(\sqrt{1+cN})^{\gamma+1}} \cdot \frac{K_{\gamma+1}(b\sqrt{1+cN})}{K_{\gamma+1}(b)} \quad (4)$$

The total active vocabulary V is given by $\lim_{N \rightarrow \infty} n_N$, which can be shown to be

$$V = \frac{2}{bc} \cdot \frac{K_{\gamma}(b)}{K_{\gamma+1}(b)} \quad (5)$$

The variances of $\hat{n}_N$ and $\hat{f}_{1,N}$, and the covariance between the two functions were derived by Sichel (1986a) for the special case $\gamma = -\frac{1}{2}$. This is now extended to the general case where $\gamma \neq -\frac{1}{2}$. Writing n_N^* and $f_{1,N}^*$ for the functions n_N and $f_{1,N}$ with c replaced by $2c$:

$$\text{var}(\hat{n}_N) = 2n_N^* - n_N - \frac{1}{N} \cdot f_{1,N}^2 \quad (6)$$

$$\text{var}(\hat{f}_{1,N}) = f_{1,N} - \frac{bcN}{2\sqrt{1+2cN}} \left[\frac{K_{\gamma}(b\sqrt{1+2cN})}{K_{\gamma+1}(b\sqrt{1+2cN})} + \frac{2(\gamma+1)}{b\sqrt{1+2cN}} \right] f_{1,N}^*$$

$$- \frac{1}{N} \left\{ 1 - \frac{bcN}{2\sqrt{1+cN}} \left[\frac{K_{\gamma}(b\sqrt{1+cN})}{K_{\gamma+1}(b\sqrt{1+cN})} + \frac{2(\gamma+1)}{b\sqrt{1+cN}} \right] \right\}^2 f_{1,N}^2 \quad (7)$$

$$\text{and } \text{cov}(\hat{f}_{1,N}, \hat{n}_N) = f_{1,N}^* - \frac{1}{N} \left\{ 1 - \frac{bcN}{2\sqrt{1+cN}} \left[\frac{K_{\gamma}(b\sqrt{1+cN})}{K_{\gamma+1}(b\sqrt{1+cN})} + \frac{2(\gamma+1)}{b\sqrt{1+cN}} \right] \right\} f_{1,N}^2 \quad (8)$$

For more detail see Price (1992). Chistyakov and Viktorova (1965) have shown that for $N \rightarrow \infty$ and $\log \left(\frac{N}{V} \right)$ bounded, the limiting distributions for all the $\hat{f}_{r,N}$ are asymptotically normal. Hence confidence limits can be calculated for the fitted functions $\tilde{n}_N$ and $\tilde{f}_{1,N}$.

The Sichel distribution has been used successfully to fit a number of datasets by the 'minimum χ^2 ' method (*e.g.* Sichel, 1992a, 1992b), and Burrell and Fenton (1993) outline a procedure for fitting the zero-truncated distribution by maximum likelihood. However, it is often the case that the full observed frequency distribution is not available, but only pairs of observations $(N, \hat{n}_N)$ and possibly $(N, \hat{f}_{1,N})$ at different levels of N . The problem is estimation of the parameters b , c and γ from the data and since the functions n_N and $f_{1,N}$ are highly nonlinear, this must be done by a nonlinear least-squares method. As a matter of convenience, formulae that follow will be set out in terms of n_N .

4. Parameter Estimation

Writing θ_1 for b , θ_2 for c , and θ_3 for γ , we denote the vector of parameters by $\underline{\theta} = (b \ c \ \gamma)^T$ (where ' $\top$ ' represents the transpose function). If there are m pairs of observations, the parameter estimates are taken as those that minimize the sum of squares S where

$$S(\underline{\theta}) = \sum_{i=1}^m r_i^2(\underline{\theta}); \quad r_i(\underline{\theta}) = [\hat{n}_{N_i} - n_{N_i}(\underline{\theta})] \quad \text{for } b > 0 \text{ and } c > 0. \quad (9)$$

Clearly this requires recourse to a numerical procedure.

Derivative-based methods generally find the minimum of S by stepping from a particular value of $\underline{\theta}$, $\underline{\theta}_{cur}$ say, to the next value, $\underline{\theta}_{nex}$ say, using

some algorithm that incorporates Taylor expansions. Two well-known iterative procedures of this type are the 'inverse-Hessian method' and the method of 'steepest descent', both of which are described in many texts on non-linear optimisation. The *Levenberg-Marquardt* algorithm (Marquardt, 1963) varies smoothly between these two methods by defining

$$\xi_k = -\frac{1}{2} \cdot \frac{\partial S}{\partial \theta_k}, \quad \eta_{kl} = \frac{1}{2} \cdot \frac{\partial^2 S}{\partial \theta_l \partial \theta_k}, \quad (10)$$

$$\eta'_{kk} = \eta_{kk} (1 + \lambda), \quad \eta'_{kl} = \eta_{kl}, \quad (k \neq l) \quad (11)$$

and solving the set of equations $((\eta')) \underline{\delta \theta} = \underline{\xi}$ where $\underline{\delta \theta} = \underline{\theta}_{nex} - \underline{\theta}_{cur}$, and λ is set in an essentially *ad hoc* manner, which works well in practice (see Press *et al.*, 1986, p. 684): it is initialised to a reasonably small value, for instance 0.001. At each step it is either increased by a fairly substantial factor (say 10) or decreased by the factor, according to whether the current sum of squares has increased or decreased.

From (9) we obtain

$$\xi_k = \sum_{i=1}^m [\hat{n}_{N_i} - n_{N_i}(\underline{\theta})] \frac{\partial n_{N_i}(\underline{\theta})}{\partial \theta_k} \quad (12)$$

$$\text{and } \eta_{kl} = \sum_{i=1}^m \left\{ \frac{\partial n_{N_i}(\underline{\theta})}{\partial \theta_l} \cdot \frac{\partial n_{N_i}(\underline{\theta})}{\partial \theta_k} - [\hat{n}_{N_i} - n_{N_i}(\underline{\theta})] \frac{\partial^2 n_{N_i}(\underline{\theta})}{\partial \theta_l \partial \theta_k} \right\}. \quad (13)$$

Many standard methods (including the Levenberg-Marquardt algorithm in its original form) proceed by ignoring the second term in the definition of η_{kl} in (13), the rationale being that the error terms $[\hat{n}_{N_i} - n_{N_i}(\underline{\theta})]$ should be random and uncorrelated, so when summed should produce a "small" contribution. This has obvious advantages if the second derivatives are difficult

(or impossible) to compute. For the purposes of the present study this second term was included since it was sometimes necessary for achieving convergence.

Starting guesses for the algorithm were obtained initially by a 'grid search' and then by implementation of a non-derivative method (the 'downhill simplex method' based on an algorithm due to Nelder and Mead, 1965). This method proceeds by constructing a simplex and moving downhill in a straightforward manner by taking a series of steps, most of which move the point of the simplex where the function is largest to a reflected point on its opposite face where the function value is lower.

Calculation of (1) and (4) require numerical evaluation of the modified Bessel function of the second kind. A FORTRAN program based mostly on an algorithm devised by Temme (1975) was used for this purpose. Implementation of a derivative-based method also requires an explicit form for the derivatives in (12) and (13). Partial differentiation of the functions n_N and $f_{1,N}$ results in a collection of lengthy expressions, which can be simplified to some extent by writing them in terms of the original functions (this also makes the task of finding second derivatives slightly easier). The expressions can be found in the Appendix.

Evaluation of these derivatives in turn requires numerical evaluation of the derivative with respect to ν of $K_\nu(z)$. An integral representation of $K_\nu(z)$ is $\int_0^\infty \cosh(\nu u) e^{-z \cosh u} du$,

$$\text{hence } \frac{\partial}{\partial \nu} K_\nu(z) = \int_0^\infty u \sinh(\nu u) e^{-z \cosh u} du \quad (14)$$

$$\text{and } \frac{\partial^2}{\partial \nu^2} K_\nu(z) = \int_0^\infty u^2 \cosh(\nu u) e^{-z \cosh u} du \quad (15)$$

which can then be computed by numerical integration.

5. Application to datasets

Sichel (1986b) expresses the belief that “most, but not all, observed word frequency distributions are well fitted by assuming $\gamma = -\frac{1}{2}$ *a priori*” (thereby simplifying the related mathematics enormously). However, the present work has established beyond any doubt that greatly improved fits can in fact be obtained by allowing γ to be estimated from the data. Once this has been achieved an estimate of the total active vocabulary can be obtained from (5).²

The curves (1) and (4) were fitted to 31 and 7 datasets respectively. As a result of space constraints, only a limited number of results can be presented so two examples (one with both type-token and *hapax legomena* data) have been chosen. More information regarding the fitted functions is available from the author on request. For the purpose of assessing the goodness-of-fit of a particular curve, the criterion of the mean absolute deviation (m.a.d.) not exceeding 10 (effectively a rule of thumb), and consequently the residual sum of squares (r.s.s.) not exceeding $100 \times \text{no. of observations}$, was used. In terms of this criterion some extremely promising results were obtained.

Figure 1 consists of data from chapter LV (the ‘*Tempest*’) of Charles Dickens’ *David Copperfield* (Thomson & Thompson, 1915). The values used in the fitting procedure are means of repeated observations at various levels of N , so this dataset provides a good example for showing confidence limits. According to the theory since there are 45 observations there should be $0.05 \times 45 = 2.25$ observations falling outside of the limits, and in fact there

² It must be stressed that the estimate of total vocabulary is highly dependent on how the data is collected and grouped. The question of how ‘distinct words’ should be defined is clearly a subjective one, and has to be addressed with the aid of a linguist. This point is expanded upon in Price (1992).

are exactly 2 for which this happens. The Sichel-model fit reduced the r.s.s. obtained by Thomson and Thompson from ≈ 564 (which was in any event remarkably good for a dataset of 16 observations) to ≈ 97 , and the m.a.d. from ≈ 4.4 to ≈ 2.9 .

Figures 2 and 3 comprise data from '*L'Illusion Comique*' by Pierre Corneille (Muller, 1964b, Table XI p. 64).

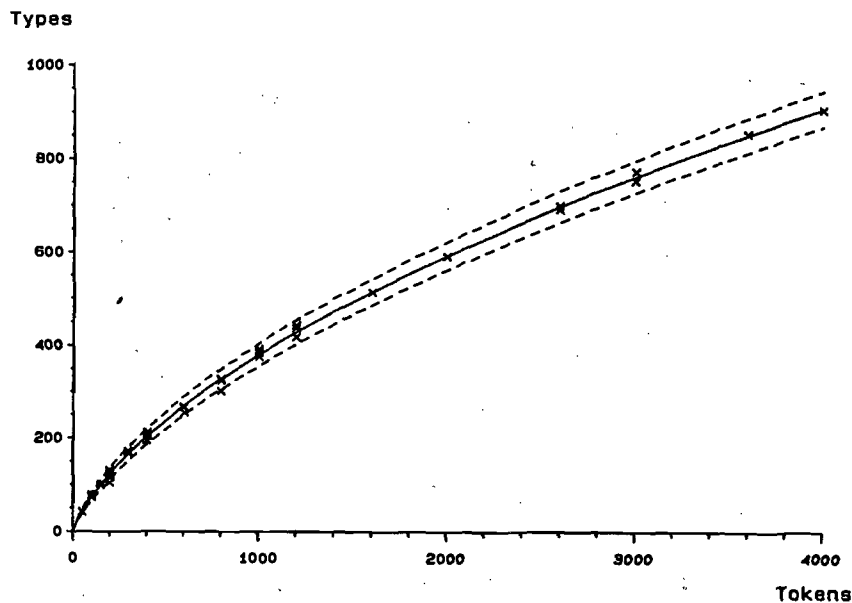


Figure 1. David Copperfield (n_N) with 95% confidence limits: $\hat{b} = 0.02039$; $\hat{c} = 0.03724$; $\hat{\gamma} = -0.6748$; $\hat{V} = 6\,864$; $r.s.s. = 197.30$; m.a.d. 2.8960.

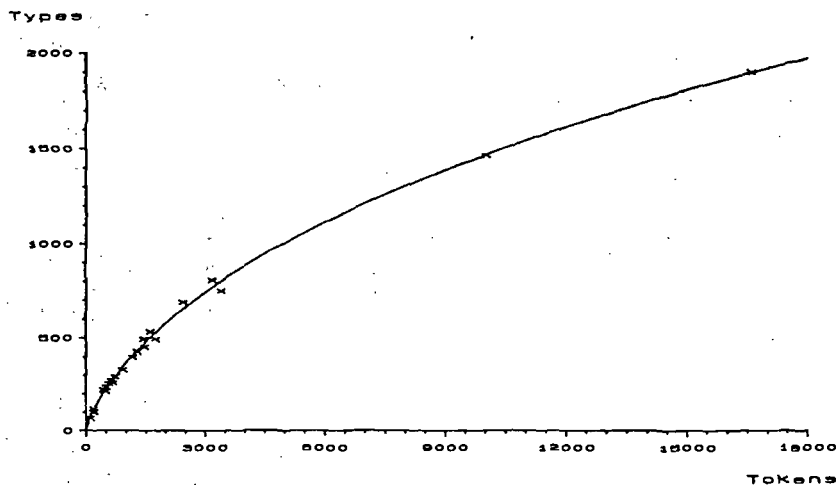


Figure 2. L'Illusion Comique (n_N): $\hat{b} = 0.02350$; $\hat{c} = 0.07712$;
 $\hat{\gamma} = -0.7781$; $\hat{V} = 4600$; $r.s.s. = 9881$ $m.a.d. = 15.817$

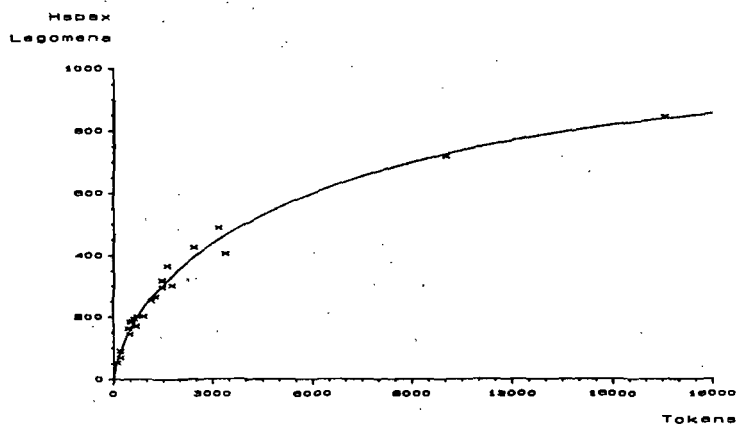


Figure 3. L'Illusion Comique ($f_{1,N}$): $\hat{b} = 0.02738$; $\hat{c} = 0.05306$;
 $\hat{\gamma} = -0.7334$; $\hat{V} = 4344$; $r.s.s. = 12705$; $m.a.d. = 18.889$

6. Extensions

The methods set out above are applicable in the 'stationary' case which occurs when the samples being drawn are from a vocabulary that remains constant. The 'non-stationary' case arises either when the writer moves from one subject to another, or when the active vocabulary of the author actually increases over a period of time. This can be observed when the speech of a young child is recorded over a period of several weeks. In the non-stationary case, if the above methods are applied and the estimated type-token parameters are used for calculating the expected number of *hapax legomena* (or *vice versa*), incorrect results are obtained. Thus a way must be found of incorporating more information from the underlying distribution into the estimation process to form a connection between the type-token characteristic and *hapax legomena* function. This point is illustrated in figures 4 and 5. The data in these figures is made up of recorded speech of a child (Kerry) aged $2\frac{1}{2}$ which was collected over a consecutive four-week period (Kerr, 1980)³.

The first two population moments about the origin for the zero-truncated GIGP were derived by Sichel (1986b). If (2) and (3) are reparameterised with $\alpha_N = b\sqrt{1 + cN}$, $\alpha_N > 0$, they are given (after some re-arrangement) by

³ A process of augmentation was used. This involves counting a certain number of tokens and noting the associated number of types (or *hapax legomena*, etc.), then obtaining the next set of tokens and augmenting the previous sample with these, hence creating a larger sample - and so forth. This technique is particularly useful when the writing or speech of a child is recorded, it being difficult (or impossible) to observe a sufficient number of tokens for the sample to be collected in any other manner. It does, however, cause a correlation to be introduced between successive data points. It should be noted that the theory that has been developed thus far pertains to independent observations, and although excellent fits have been obtained for augmented data sets this is a question that should be addressed in future research, since the serial correlation tends to result in oscillations about the fitted curve.

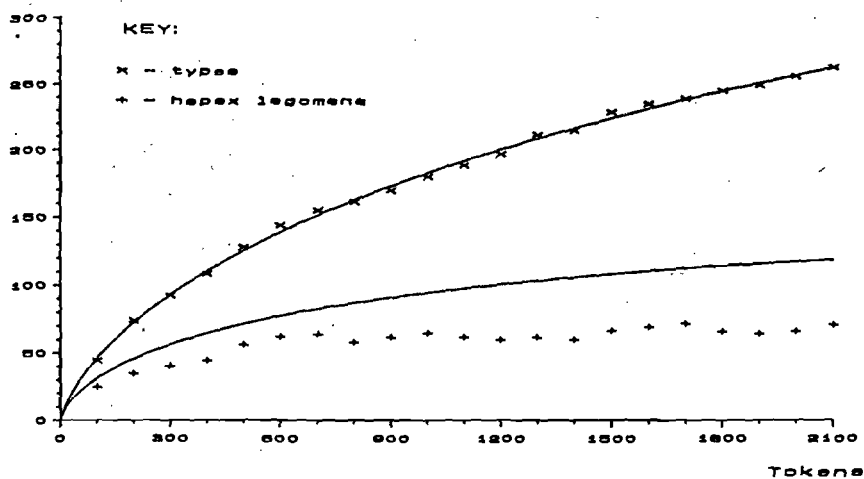


Figure 4. Child, Kerry (Estimates based on only n_N): $\hat{b} = 0.03844$;
 $\hat{c} = 0.1914$; $\hat{\gamma} = -0.6904$

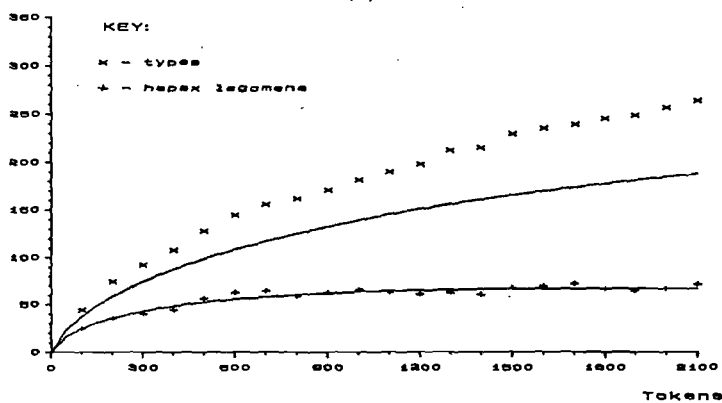


Figure 5. Child, Kerry (Estimates based on only $f_{1,N}$): $\hat{b} = 0.03860$;
 $\hat{c} = 0.6775$; $\hat{\gamma} = -0.8155$

$$\phi(1|N) = \frac{(\alpha^2 - b^2)}{2} \cdot \frac{\alpha^{-(\gamma+1)} K_{\gamma+1}(\alpha)}{[b^{-\gamma} K_{\gamma}(b) - \alpha^{-\gamma} K_{\gamma}(\alpha)]} \quad (16)$$

$$\text{and} \quad \frac{E(r|N)}{\phi(1|N)} = \left(\frac{\alpha}{b}\right)^{\gamma+1} \cdot \frac{K_{\gamma+1}(b)}{K_{\gamma+1}(\alpha)} \quad (17)$$

(where the subscript 'N' has been dropped from α as a matter of convenience).

If we let $\theta_N = \frac{cN}{1 + cN}$ (where now $0 < \theta_N < 1$) then at each level of N , $\theta_N = 1 - \left(\frac{b_N}{\alpha_N}\right)$ and c_N can be calculated from $c_N = \frac{\theta_N}{(1 - \theta_N)N}$. The non-stationarity causes b and c themselves to become dependent on N , as distinct from the theoretical case where stationarity holds and these parameters are invariant.

Since in the present application we have only triplets of observations $(N, \hat{n}_N, \hat{f}_{1,N})$ available at various levels of N (and not the full observed frequency distribution), it is not possible to estimate α and b by (for example) setting $\phi(1|N) = \frac{\hat{f}_{1,N}}{\hat{n}_N}$ and $\frac{E(r|N)}{\phi(1|N)} = \frac{N}{\hat{f}_{1,N}}$, and then applying a 'minimum χ^2 ' method to find γ . Instead an (essentially *ad hoc*) algorithm has been developed. By defining

$$g_1(\alpha, b) = \frac{N}{\hat{f}_{1,N}} - \left(\frac{\alpha}{b}\right)^{\gamma+1} \cdot \frac{K_{\gamma+1}(b)}{K_{\gamma+1}(\alpha)} \quad (18)$$

$$g_2(\alpha, b) = \frac{\hat{n}_N}{\hat{f}_{1,N}} \cdot \frac{(\alpha^2 - b^2)}{2} \cdot \frac{K_{\gamma+1}(\alpha)}{\alpha^{\gamma+1}} + \frac{K_{\gamma}(\alpha)}{\alpha^{\gamma}} - \frac{K_{\gamma}(b)}{b^{\gamma}} \quad (19)$$

the *Newton-Raphson* method can be used to solve for simultaneous zeros. In this case there are two possibilities :

- (1) The function g_1 can be considered as a function of b only and g_2 as a function of α only, and a 'one-dimensional' method employed as follows :

- (i) α and b are initialised to α_0 and b_0 which are taken as α_{cur} and b_{cur} on the first iteration;
 - (ii) a solution for b is found using the Newton-Raphson equation $b_{nex} = b_{cur} - \frac{g_1(b_{cur})}{\frac{\partial}{\partial b} g_1(b_{cur})}$ with α_{cur} being substituted in (18);
 - (iii) α is updated with the formula $\alpha_{nex} = \alpha_{cur} - \frac{g_1(\alpha_{cur})}{\frac{\partial}{\partial \alpha} g_1(\alpha_{cur})}$ where the solution for b found in (ii) is now used in (19). α is checked for convergence and if this condition has been fulfilled a solution for b is found using the final value of α ; if not, the process is continued at (ii).
- (2) The other option is for the functions g_1 and g_2 to be solved simultaneously for zero using a 'two-dimensional' method as follows :
- (i) α and b are initialised to α_0 and b_0 as in (1) above
 - (ii) the Newton-Raphson equations $\frac{\partial g_i}{\partial \theta_1} \frac{\delta \theta_1}{\partial \theta_2} + \frac{\partial g_i}{\partial \theta_2} \frac{\delta \theta_2}{\partial \theta_2} = -g_i(\underline{\theta})$, $i = 1, 2$ (where $\underline{\theta}$, the vector of parameters, is now defined by $\underline{\theta} = (\theta_1 \theta_2)^\top \equiv (\alpha \ b)^\top$), are solved for the increments $\delta \underline{\theta}$ which are then added to the current values of $\underline{\theta}$ and the process is iterated to convergence.

The first method is useful for obtaining starting values α_0 and b_0 for the second algorithm which obviously gives more accurate solutions. This procedure can be applied to a set of triplets $(N, \hat{n}_N, \hat{f}_{1,N})$ through a range of token values to obtain estimates for α_N and b_N if γ has been fixed beforehand.

For each progression of $\hat{b}$'s found by setting γ at a particular value, a corresponding progression of $\hat{c}$'s exists which can be modelled by the function

$$c_N = a_0 + a_1 e^{-a_2 N} \quad (20)$$

where a_0 , a_1 and a_2 are constants that can be found very simply by least-squares estimation. Starting values for the constants are readily available

since the initial guess for $\hat{a}_0$ (the asymptote) can be set to a value slightly lower than the lowest 'observed'⁴ $\hat{c}_N$, resulting in a linearisable function (*i.e.* $c_N - a_0 = a_1 e^{-a_2 N}$). Thus linear regression can be used to find starting values for $\hat{a}_1$ and $\hat{a}_2$. This provides a method for estimating the parameters :

- Step 1.** Adjust γ through a range of values, finding the $\hat{b}_N$'s and $\hat{c}_N$'s for each one.
- Step 2.** Establish the value of γ for which the progression of $\hat{b}_N$'s is most stable.
- Step 3.** Take this value of γ to be its estimate $\tilde{\gamma}$; take the average of the $\hat{b}_N$'s to be the estimate $\tilde{b}$ of b , and fit the corresponding progression of $\hat{c}_N$'s according to (20) which gives $\tilde{c}_N$. Then expected values for both n_N and $f_{1,N}$ can be computed, with their associated r.s.s.'s and m.a.d.'s.

Quantities such as $\tilde{V}_N$ (now monotonically increasing with N) and - when data sets with independent observations are available - the variances of $\hat{n}_N$ and $\hat{f}_{1,N}$ as well as the covariance between the two functions, can be computed. The *overall* limiting vocabulary can also be evaluated, since $c_\infty = a_0$ and so from (5),

$$V_{lim} = \frac{2}{ba_0} \cdot \frac{K_\gamma(b)}{K_{\gamma+1}(b)}$$

The algorithm outlined above worked very well in the examples considered. Figure 6 illustrates the improvement over the fits previously obtained for this data.

⁴ This word is used in a somewhat loose sense since the $\hat{c}_N$ are actually estimated from the data. The convention used for denoting estimates is for so-called observed b 's and c 's to be written with a 'hat', and final estimates obtained from these to be written with a 'tilde'.

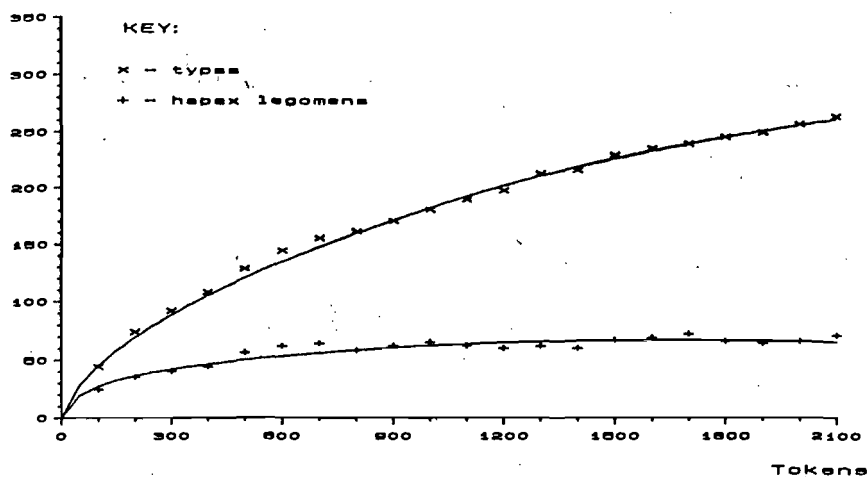


Figure 6. Child, Kerry (Estimates based on both n_N and $f_{1,N}$): $\hat{b} = 0.1812$; $\hat{a}_0 = 0.06873$; $\hat{a}_1 = 0.09703$; $\hat{a}_2 = 1.652 \times 10^{-3}$; $\hat{\gamma} = -0.85$; $V_{lim} = 343$

During the fitting procedure it became apparent that including low token counts (for instance anything up to two or three hundred) had a destabilizing effect on the parameter estimates and better fits could in fact be obtained by omitting the first few observations. The estimates were thus computed by successively removing the lowest observation from the dataset, repeating the procedure at each stage⁵. Since this work is still in the exploratory stage, no procedure has yet been outlined for computing the standard error of the estimates.

Table 1. Goodness-of-fit measures for child data, Kerry (Estimates based on both n_N and $f_{1,N}$)

function	rss.	m.a.d.
$n_N :$	331.76	3.0693
$f_{1,N} :$	356.94	3.3151

7. Conclusions

The results obtained from fitting the type-token characteristic based on the generalised inverse Gaussian-Poisson distribution were extremely encouraging and show that Sichel's model is not only attractive in theory, but also works well in practice. The new procedure proposed for addressing the 'non-stationary' case where the model parameters themselves become functions of the number of tokens N , has yielded promising results.

⁵ Clearly, as more observations are excluded there must come a point where not enough information is being utilised for effective estimation to take place - so for example it might be found that the 'best' fit can be obtained by omitting only the first observation or the first two observations.

8. Acknowledgements

I am grateful to the late Professor H.S. Sichel for introducing me to this subject and for extensive guidance at all stages of the work upon which this paper is based.

I would also like to thank Mr. N. Sennett (formerly of the Department of Statistics and Actuarial Science, University of the Witwatersrand), for his FORTRAN program which evaluates the modified Bessel function of the second kind.

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10. Appendix

APPENDIX A: Jacobian and Hessian for n_N and $f_{1,N}$

First Derivatives

$$\frac{\partial n_N}{\partial b} = \left[\frac{K_\gamma(b)}{K_{\gamma+1}(b)} + \frac{2\gamma}{b} \right] n_N + \frac{2}{bc} \left[\frac{(1+cN)}{N} f_{1,N} - 1 \right]$$

$$\frac{\partial n_N}{\partial c} = \frac{1}{c} [f_{1,N} - n_N]$$

$$\begin{aligned} \frac{\partial n_N}{\partial \gamma} &= [G_\gamma(b\sqrt{1+cN}) - \ln \sqrt{1+cN} - G_{\gamma+1}(b)] n_N \\ &\quad - \frac{2}{bc} \cdot \frac{K_\gamma(b)}{K_{\gamma+1}(b)} [G_\gamma(\sqrt{1+cN}) - \ln \sqrt{1+cN} - G_\gamma(b)] \end{aligned}$$

$$\frac{\partial f_{1,N}}{\partial b} = \frac{K_\gamma(b)}{K_{\gamma+1}(b)} [f_{1,N} - N] + \frac{bcN}{2} n_N$$

$$\frac{\partial f_{1,N}}{\partial c} = \frac{-N}{\sqrt{1+cN}} \left[\frac{b}{2} \cdot \frac{K_\gamma(b\sqrt{1+cN})}{K_{\gamma+1}(b\sqrt{1+cN})} + \frac{(\gamma+1)}{\sqrt{1+cN}} \right] f_{1,N}$$

$$\frac{\partial f_{1,N}}{\partial \gamma} = [G_{\gamma+1}(b\sqrt{1+cN}) - \ln \sqrt{1+cN} - G_{\gamma+1}(b)] f_{1,N}$$

where $G_\nu(z) = \frac{K_\nu^*(z)}{K_\nu(z)}$; $K_\nu^*(z) = \frac{\partial}{\partial \nu} K_\nu(z)$

Second Derivatives

$$\begin{aligned} \frac{\partial^2 n_N}{\partial b^2} = & \left\{ 2 \left[\frac{K_\gamma(b)}{K_{\gamma+1}(b)} + \frac{2\gamma}{b} \right] \left[\frac{K_\gamma(b)}{K_{\gamma+1}(b)} + \frac{(\gamma + \frac{1}{2})}{b} \right] - \frac{4\gamma}{b^2} + cN \right\} n_N \\ & + \frac{4}{bc} \left[\frac{K_\gamma(b)}{K_{\gamma+1}(b)} + \frac{(\gamma - \frac{1}{2})}{b} \right] \left[\frac{(1 + cN)}{N} f_{1,N} - 1 \right] \\ & - \frac{2N}{b} \cdot \frac{K_\gamma(b)}{K_{\gamma+1}(b)} \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 n_N}{\partial c^2} = & \frac{2}{c^2} n_N - \frac{N}{c} \left[\frac{b}{2\sqrt{1 + cN}} \cdot \frac{K_\gamma(b\sqrt{1 + cN})}{K_{\gamma+1}(b\sqrt{1 + cN})} \right. \\ & \left. + \frac{(\gamma + 1)}{(1 + cN)} + \frac{2}{cN} \right] f_{1,N} \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 n_N}{\partial \gamma^2} = & \left\{ G_\gamma^*(b\sqrt{1 + cN}) - G_{\gamma+1}^*(b) + (\ln \sqrt{1 + cN})^2 \right. \\ & - 2 [G_\gamma(b\sqrt{1 + cN}) - G_{\gamma+1}(b)] [G_{\gamma+1}(b) + \ln \sqrt{1 + cN}] \} n_N \\ & - \frac{2}{bc} \cdot \frac{K_\gamma(b)}{K_{\gamma+1}(b)} \left\{ G_\gamma^*(b\sqrt{1 + cN}) - G_\gamma^*(b) + (\ln \sqrt{1 + cN})^2 \right. \\ & - 2G_{\gamma+1}(b) [G_\gamma(b\sqrt{1 + cN}) - G_\gamma(b)] \\ & \left. - \ln(1 + cN) [G_\gamma(b\sqrt{1 + cN}) - G_{\gamma+1}(b)] \right\} \end{aligned}$$

$$\text{where } G_\nu^*(z) = \frac{K_\nu^{**}(z)}{K_\nu(z)}; \quad K_\nu^{**}(z) = \frac{\partial^2}{\partial \nu^2} K_\nu(z)$$

$$\frac{\partial^2 f_{1,N}}{\partial b^2} = \left[\frac{K_\gamma(b)}{K_{\gamma+1}(b)} + \frac{(\gamma + \frac{1}{2})}{b} \right] \left\{ 2 \frac{K_\gamma(b)}{K_{\gamma+1}(b)} (f_{1,N} - N) + b c N n_N \right\} + c N f_{1,N}$$

$$\frac{\partial^2 f_{1,N}}{\partial c^2} = \frac{b N^2}{2(1+cN)} \left\{ \frac{b}{2} + \frac{(\gamma+2)}{\sqrt{1+cN}} \left[\frac{K_\gamma(b\sqrt{1+cN})}{K_{\gamma+1}(b\sqrt{1+cN})} + \frac{2(\gamma+1)}{b\sqrt{1+cN}} \right] \right\} f_{1,N}$$

$$\begin{aligned} \frac{\partial^2 f_{1,N}}{\partial \gamma^2} = & \left\{ G_{\gamma+1}^*(b\sqrt{1+cN}) - G_{\gamma+1}^*(b) + (\ln \sqrt{1+cN})^2 \right. \\ & + 2 [G_{\gamma+1}(b) - G_{\gamma+1}(b\sqrt{1+cN})] \\ & \left. [G_{\gamma+1}(b) + \ln \sqrt{1+cN}] \right\} f_{1,N} \end{aligned}$$

Cross-Derivatives

$$\begin{aligned} \frac{\partial^2 n_N}{\partial b \partial c} = & \frac{1}{c} \cdot \frac{K_\gamma(b)}{K_{\gamma+1}(b)} [f_{1,N} - n_N - N] + \frac{bN}{2} n_N + \frac{2}{bc} \\ & \cdot \left[\frac{1}{c} - \gamma n_N - \frac{(1+cN)}{cN} f_{1,N} \right] \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 n_N}{\partial b \partial \gamma} = & \left\{ \frac{K_\gamma(b)}{K_{\gamma+1}(b)} [G_\gamma(b) - G_{\gamma+1}(b)] + \frac{2}{b} \right\} n_N \\ & + \frac{2}{bc} \cdot \frac{(1+cN)}{N} [G_{\gamma+1}(b\sqrt{1+cN}) - \ln \sqrt{1+cN} - G_{\gamma+1}(b)] \\ & \cdot f_{1,N} + \left[\frac{K_\gamma(b)}{K_{\gamma+1}(b)} + \frac{2\gamma}{b} \right] \{ [G_\gamma(b\sqrt{1+cN}) - \ln \sqrt{1+cN}] \\ & \cdot [n_N - \frac{2}{bc} \cdot \frac{K_\gamma(b)}{K_{\gamma+1}(b)}] - G_{\gamma+1}(b) n_N + \frac{2}{bc} \cdot \frac{K_\gamma(b)}{K_{\gamma+1}(b)} G_\gamma(b) \} \end{aligned}$$

$$\begin{aligned}
\frac{\partial^2 n_N}{\partial c \partial \gamma} &= \frac{1}{c} \{ G_{\gamma+1}(b\sqrt{1+cN}) f_{1,N} - G_\gamma(b\sqrt{1+cN}) \\
&\quad \left[n_N - \frac{2}{bc} \cdot \frac{K_\gamma(b)}{K_{\gamma+1}(b)} \right] + [\ln \sqrt{1+cN} + G_{\gamma+1}(b)] (n_N - f_{1,N}) \\
&\quad - \frac{2}{bc} \cdot \frac{K_\gamma(b)}{K_{\gamma+1}(b)} [\ln \sqrt{1+cN} + G_\gamma(b)] \} \\
\frac{\partial^2 f_{1,N}}{\partial b \partial c} &= N \left\{ \frac{b}{2} - \frac{K_\gamma(b)}{\sqrt{1+cN} K_{\gamma+1}(b)} \left[\frac{b}{2} \cdot \frac{K_\gamma(b\sqrt{1+cN})}{K_{\gamma+1}(b\sqrt{1+cN})} \right. \right. \\
&\quad \left. \left. + \frac{(\gamma+1)}{\sqrt{1+cN}} \right] \right\} f_{1,N} \\
\frac{\partial^2 f_{1,N}}{\partial b \partial \gamma} &= \frac{K_\gamma(b)}{K_{\gamma+1}(b)} [G_{\gamma+1}(b\sqrt{1+cN}) - \ln \sqrt{1+cN} + G_\gamma(b) \\
&\quad - 2G_{\gamma+1}(b)] f_{1,N} + N [G_\gamma(b\sqrt{1+cN}) - \ln \sqrt{1+cN} - G_{\gamma+1}(b)] \\
&\quad \cdot \left\{ \frac{bc}{2} n_N - \frac{K_\gamma(b)}{K_{\gamma+1}(b)} \right\} \\
\frac{\partial^2 f_{1,N}}{\partial c \partial \gamma} &= \frac{-N}{(1+cN)} \{ (\gamma+1) [G_{\gamma+1}(b\sqrt{1+cN}) - \ln \sqrt{1+cN} \\
&\quad - G_{\gamma+1}(b)] + 1 + \frac{b\sqrt{1+cN}}{2} \cdot \frac{K_\gamma(b\sqrt{1+cN})}{K_{\gamma+1}(b\sqrt{1+cN})} \\
&\quad \cdot [G_\gamma(b\sqrt{1+cN}) - \ln \sqrt{1+cN} - G_{\gamma+1}(b)] \} f_{1,N}
\end{aligned}$$